

Class #7 - Transformation

1. Suppose $b \in \text{Span}\{a_1, a_2\}$. Then _____ for _____ $c_i \in R$. Let

$$A = \begin{bmatrix} a_1 & a_2 \end{bmatrix}, \quad x = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}.$$

Then, we can write the vector equation as a matrix equation:

$$Ax = b$$

where Ax is defined as _____.

Note: The vector equation _____, the matrix equation _____, and the system of linear equations with the augmented matrix _____ are all _____.

2. We can think of Matrix-vector multiplication as a *transformation*:

Take any $x \in \mathbf{R}^2$. $x =$ _____.

Want to transform this vector into a vector in $\mathbf{R}^4$: pick a matrix A of the size _____ so that

$$Ax = \text{_____}.$$

(The set of all $m \times n$ matrices (with real entries) is denoted as $\mathbf{R}^{m \times n}$.)

3. Example: Let $T : \mathbf{R}^2 \rightarrow \mathbf{R}^4$ be defined by

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

(a) $T(x_1, x_2) = (\text{_____, } \text{_____, } 0, \text{_____})$.

(b) Find the image of $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

(c) Find an $x \in \mathbf{R}^2$ whose image under T is $(0, 0, 0, 0)$.

(d) Is there more than one $x \in \mathbf{R}^2$ so that $T(x) = 0$?

(e) Can $\begin{bmatrix} 3 \\ 3 \\ 1 \\ 6 \end{bmatrix}$ be an image of some $x \in \mathbf{R}^2$?

(f) Is $(3, 3, 1, 6)$ in the Range of T ?

(g) Find a vector in $\mathbf{R}^4$ which is in $\text{Range}(T)$.

4. A transformation $T : \mathbf{R}^m \rightarrow \mathbf{R}^n$ is a *linear transformation* if

(i) $T(u + v) = \underline{\hspace{2cm}}$, and

(ii) $T(cu) = \underline{\hspace{2cm}}$.

5. Examples:

(a) $T : \mathbf{R}^m \rightarrow \mathbf{R}^n$ where $T(x) = Ax$:

(b) $T : \mathbf{R}^2 \rightarrow \mathbf{R}^3$ where $T(x) = (x_1, x_2, 5x_1)$

(c) $T : \mathbf{R}^2 \rightarrow \mathbf{R}^2$ where $T(x) = (x_1 + 3, x_2)$